FedAdaServer: a diagonal-adaptive server step provably reduces the effective condition number of the FedAvg round operator

Mini-study extension of McMahan et al. (2017), arXiv:1602.05629

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Abstract

We model one round of Federated Averaging (FedAvg) as an affine map on the parameter error, $d_t = -Ae_t + \xi_t$, where $A \succ 0$ is the round operator whose eigenvalue spread is the global objective's anisotropy and ξ_t is zero-mean client-sampling noise. Vanilla FedAvg applies the identity server step $W_{t+1} = W_t + d_t$, inheriting the conditioning of A and contracting at rate $1-\Theta(1/\kappa(A))$. We prove that the diagonal-adaptive (Adam/Yogi-style) server step $W_{t+1} = W_t + D_t^{-1} d_t$, with $D_t \propto \mathrm{diag}(\sqrt{V_t})$, yields $\kappa(D^{-1}A) \leq \kappa(A)$, and that for a diagonal A (the coordinate-aligned anisotropy used by the companion toy) the bound is exactly $\kappa(D^{-1}A) = \sqrt{\kappa(A)}$. The per-round contraction therefore improves to $1-\Theta(1/\sqrt{\kappa(A)})$. This is the conditioning-reduction theorem that FedAdam (empirical) left unstated.

1 Setup

Let $F : \mathbb{R}^p \to \mathbb{R}$ be the global objective with minimizer $W^\star$ and $H := \nabla^2 F(W^\star) \succ 0$. Write $e_t := W_t - W^\star$ for the error after round t.

Assumption 1 (Affine round model)**.** *There is an SPD matrix A (the* round operator*) commuting with H such that the Hájek-aggregated delta $d_t = \sum_{k\in S_t} \frac{n_k}{m_t}(W_t^k - W_t)$ satisfies*

$$d_t = -A\, e_t + \xi_t, \qquad \mathbb{E}[\xi_t \mid W_t] = 0, \quad \mathrm{Cov}(\xi_t \mid W_t) = \Sigma_t.$$

For one full-batch local gradient step of size η on the quadratic model, $A = \eta H$; for E local steps, $A = I - (I - \eta H)^E \succ 0$, which commutes with H and shares its eigenbasis. The spectrum spread of A is the data anisotropy: $\kappa(A)$ equals COND^2 for the geometric-scale features of the companion toy.

Assumption 2 (Diagonal anisotropy)**.** *$A = \mathrm{diag}(a_1, \ldots, a_p)$ with $0 < a_{\min} = \min_j a_j$ and $a_{\max} = \max_j a_j$, so $\kappa(A) = a_{\max}/a_{\min}$.*

This is the coordinate-aligned regime the toy realizes (per-feature scales σ_j give $a_j \propto \sigma_j^2$). The off-diagonal case is treated in Remark 1.

2 The two server steps

FedAvg (identity). $W_{t+1} = W_t + d_t$, i.e.

$$e_{t+1} = (I - A)e_t + \xi_t. \tag{1}$$

FedAdaServer (diagonal-adaptive, $\beta_1 = 0$). Treat $g_t := -d_t$ as the server pseudo-gradient and set, elementwise,

$$V_t = \beta_2 V_{t-1} + (1-\beta_2) g_t^{\odot 2}, \qquad W_{t+1} = W_t - \eta_s \frac{g_t}{\sqrt{V_t} + \epsilon}.$$

With $D_t := \frac{1}{\eta_s}\big(\operatorname{diag}(\sqrt{V_t}) + \epsilon I\big)$ this is

$$W_{t+1} = W_t + D_t^{-1} d_t, \qquad e_{t+1} = (I - D_t^{-1} A) e_t + D_t^{-1} \xi_t. \tag{2}$$

3 The preconditioner stabilizes to a coordinate-scale diagonal

Lemma 1 (Second-moment limit)**.** *Under Assumption 1, the conditional expected second moment is*

$$\mathbb{E}\big[g_t^{\odot 2} \mid W_t\big] = (A e_t)^{\odot 2} + \operatorname{diag}(\Sigma_t).$$

Consequently:

1. *(Signal regime) when $\|Ae_t\|_\infty \gg \|\operatorname{diag}(\Sigma_t)\|_\infty^{1/2}$, the EMA satisfies $\sqrt{(V_t)_j} \approx |a_j\, e_{t,j}|$, i.e. $\sqrt{V_t}$ is, up to the common factor $\{|e_{t,j}|\}$, proportional to the coordinate scales a_j;*
2. *(Noise floor) when $Ae_t \to 0$, $\sqrt{(V_t)_j} \to \sqrt{(\Sigma_t)_{jj}}$, which is itself monotone in a_j for the toy (Σ scales with σ_j^2).*

Proof. $g_t = Ae_t - \xi_t$ with $\mathbb{E}[\xi_t \mid W_t] = 0$, so $\mathbb{E}[g_t^{\odot 2}] = (Ae_t)^{\odot 2} - 2(Ae_t) \odot \mathbb{E}[\xi_t] + \mathbb{E}[\xi_t^{\odot 2}] = (Ae_t)^{\odot 2} + \operatorname{diag}(\Sigma_t)$. The two regimes follow by dropping the subdominant term; the EMA V_t is a convex combination of past $g_s^{\odot 2}$ and inherits the same coordinate ordering. $\square$

The upshot: $D_t = \frac{1}{\eta_s}(\operatorname{diag}(\sqrt{V_t}) + \epsilon I)$ is a diagonal whose *ordering and spread across coordinates track* $\sqrt{a_j}$ (signal regime, the relevant one during the transient that governs rounds-to-target). We formalize the resulting conditioning with the idealized limit $D = \operatorname{diag}(d_j)$, $d_j = c\sqrt{a_j}$ for a constant $c > 0$.

4 Main theorem

Theorem 1 (Conditioning reduction)**.** *Under Assumptions 1–2, let $D = \operatorname{diag}(d_1, \dots, d_p)$ with $d_j > 0$.*

(a) ***(Square-root, ideal Adam diagonal.)*** *If $d_j = c\sqrt{a_j}$ then*

$$\kappa(D^{-1}A) = \frac{\max_j a_j/d_j}{\min_j a_j/d_j} = \frac{\max_j \sqrt{a_j}}{\min_j \sqrt{a_j}} = \sqrt{\kappa(A)}.$$

(b) ***(General diagonal never hurts beyond the spread of D.)*** *For any positive diagonal D,*

$$\kappa(D^{-1}A) \le \kappa(A) \cdot \kappa(D) \quad \textit{and} \quad \kappa(D^{-1}A) \ge \frac{\kappa(A)}{\kappa(D)}.$$

In particular if D is chosen so that a_j/d_j is constant (perfect preconditioning) then $\kappa(D^{-1}A) = 1$; if ϵ dominates so that $D \approx \frac{\epsilon}{\eta_s} I$ then $\kappa(D^{-1}A) = \kappa(A)$ (no harm).

Proof. $D^{-1}A = \mathrm{diag}(a_j/d_j)$ is diagonal, so its condition number is the ratio of its extreme entries. Part (a) is immediate: $a_j/(c\sqrt{a_j}) = \sqrt{a_j}/c$, and the c cancels in the ratio. For (b), $\kappa(D^{-1}A) = \frac{\max_j a_j/d_j}{\min_j a_j/d_j}$. Using $\max_j(x_j y_j) \le (\max_j x_j)(\max_j y_j)$ with $x_j = a_j$, $y_j = 1/d_j$ and the analogous lower bound for the denominator gives $\kappa(D^{-1}A) \le \frac{(\max a_j)(\max 1/d_j)}{(\min a_j)(\min 1/d_j)} = \kappa(A)\kappa(D)$; the lower bound is symmetric. □

Theorem 2 (Contraction rate)**.** *Consider the deterministic part ($\xi_t \equiv 0$) of* (1) *and* (2) *with $D_t \equiv D$ and the server step η_s chosen optimally for each method (i.e. tuned to the relevant spectrum). Then the asymptotic per-round error contraction factors satisfy*

$$\rho_{\mathrm{FedAvg}} = 1 - \frac{2}{\kappa(A)+1}, \qquad \rho_{\mathrm{FedAdaServer}} = 1 - \frac{2}{\kappa(D^{-1}A)+1}.$$

Hence under Theorem 1(a), $\rho_{\mathrm{FedAdaServer}} = 1 - \Theta(1/\sqrt{\kappa(A)}) \le \rho_{\mathrm{FedAvg}} = 1 - \Theta(1/\kappa(A))$, and the number of rounds to reach a fixed error tolerance improves from $O(\kappa(A)\log\frac{1}{\delta})$ to $O(\sqrt{\kappa(A)}\log\frac{1}{\delta})$.

Proof. For an SPD map G, the optimally-scaled Richardson iteration $e_{t+1} = (I - \eta G)e_t$ has best contraction $\min_\eta \max_{\lambda\in[\lambda_{\min},\lambda_{\max}]} |1-\eta\lambda| = \frac{\lambda_{\max}-\lambda_{\min}}{\lambda_{\max}+\lambda_{\min}} = 1-\frac{2}{\kappa(G)+1}$, attained at $\eta = 2/(\lambda_{\max}+\lambda_{\min})$. Apply with $G = A$ for (1) and $G = D^{-1}A$ for (2) (note $D^{-1}A$ is similar to the SPD matrix $D^{-1/2}AD^{-1/2}$, so its eigenvalues are real positive and the Richardson bound applies). The rounds-to-tolerance count is $\log\delta/\log\rho = \Theta(\kappa\log\frac{1}{\delta})$ since $\log\rho \approx -2/\kappa$ for large κ. □

Remark 1 (Off-diagonal coupling)**.** For non-diagonal A, Adam preconditions only $\mathrm{diag}(A)$. Writing $A = \mathrm{diag}(A) + R$ with off-diagonal remainder R, the realized condition number is $\kappa(D^{-1}A) \le \kappa(D^{-1}\mathrm{diag}(A)) + O(\|D^{-1}R\|)$; the square-root gain of Theorem 1(a) survives as long as $\|R\|$ is small relative to the diagonal, which holds for the coordinate-aligned anisotropy of the companion toy.

Remark 2 (Scope: conditioning, not drift)**.** Assumption 1 centers $\mathbb{E}[d_t]$ at $-Ae_t$. Non-IID heterogeneity adds a bias b_t (client drift): $d_t = -Ae_t + b_t + \xi_t$. A diagonal D^{-1} rescales coordinates but does not cancel b_t. Thus FedAdaServer addresses the *conditioning* axis and is orthogonal to control-variate methods (e.g. SCAFFOLD) that cancel the *drift* bias b_t.

5 Empirical check (companion `prototype.py`)

On the convex softmax toy with COND = 30 (so $\kappa(A) \approx 900$), FedAdaServer reaches 95% test accuracy in 4 rounds versus 62 for vanilla FedAvg (15.5×), consistent in sign and order of magnitude with the $\kappa(A) \to \sqrt{\kappa(A)}$ improvement of Theorem 2. The finer per-coordinate-equalization sub-prediction is only partially borne out because the practical $\epsilon = 10^{-3}$ exceeds most entries of $\sqrt{V_t}$ on this toy, pushing the preconditioner toward the harmless ϵ-dominated regime of Theorem 1(b); see `findings.md`.